LEVEL 6 · PART III

Cloud (AWS)

6

Goal: Learn one cloud deeply; understand AWS building blocks well enough to design and secure a production architecture.

RECOMMENDED TIME
4-6 weeks

TOPICS
18

PRACTICE / CAPSTONE
Build CloudFront → ALB → EC2 instances → Redis → RDS → S3 (use free tier / tear down after; a billing-alarm task is mandatory).

Topics

#	Topic	What you will learn
1	EC2 **p.1141**	instances, AMIs, instance types, key pairs, user data, pricing models intro
2	Auto Scaling **p.1155**	launch templates, ASG, scaling policies, health checks, lifecycle hooks
3	Lambda **p.1167**	event-driven functions, cold starts, triggers, limits, when not to use
4	VPC and Subnets **p.1179**	CIDR planning, public/private subnets, AZs, multi-AZ design
5	Route Tables, Internet Gateway and NAT Gateway **p.1191**	how traffic leaves a subnet, NAT cost, VPC endpoints
6	Security Groups and Network ACLs **p.1203**	stateful vs stateless, SG referencing, layered defense
7	IAM **p.1217**	users, roles, policies, least privilege, instance profiles, STS, policy evaluation
8	S3 **p.1231**	buckets, policies, versioning, lifecycle, encryption, presigned URLs, static hosting
9	EBS and EFS **p.1245**	block vs file storage, snapshots, volume types, when to use which
10	RDS and Aurora **p.1261**	managed Postgres/MySQL, Multi-AZ vs read replicas, backups, Aurora architecture
11	ElastiCache **p.1273**	Redis/Memcached, caching patterns, clusters, failover

#	Topic	What you will learn
12	Route 53 **p.1285**	hosted zones, record types, routing policies, health checks, failover
13	CloudFront **p.1297**	CDN distributions, origins, caching behaviors, OAC, invalidation
14	Application Load Balancer **p.1311**	listeners, rules, target groups, health checks, TLS with ACM
15	KMS **p.1325**	keys, envelope encryption, key policies, rotation
16	Secrets Manager **p.1339**	storing/rotating secrets, app retrieval, vs SSM Parameter Store
17	CloudTrail **p.1353**	API audit logs, management vs data events, investigation queries
18	CloudWatch **p.1369**	metrics, logs, alarms, dashboards, Logs Insights, billing alarm
★	Level capstone project **p.1383**	Build CloudFront → ALB → EC2 instances → Redis → RDS → S3 (use free tier / tear down after; a billing-alarm task is mandatory).

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow



17
CloudTrail

18
CloudWatch

★
Capstone